What Are Acids and Bases?

Definitions, conjugate pairs, strength and neutralization

Name _____ Class _____ Date _____

A. Definitions

1. Define a Brønsted–Lowry acid and a Brønsted–Lowry base.

2. Explain the difference between a **strong** acid and a **concentrated** acid.

B. Acids, bases and conjugate pairs

3. For each reaction, identify the acid and the base (Brønsted–Lowry):

(a) $HNO_3 + H_2O \longrightarrow H_3O^+ + NO_3^-$

(b) $NH_3 + HCl \longrightarrow NH_4^+ + Cl^-$

(c) $HCO_3^- + OH^- \longrightarrow CO_3^{2-} + H_2O$

4. Write the **conjugate base** of: (a) HF, (b) H_2SO_4, (c) NH_4^+, (d) H_2O.

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5. Write the **conjugate acid** of: (a) NH_3, (b) OH^-, (c) H_2O, (d) CO_3^{2-}.

C. Strong and weak

6. Classify each as a strong acid, weak acid, strong base or weak base: HCl, CH_3COOH, NaOH , NH_3, HNO_3.

D. Neutralization

7. Write balanced equations for these neutralizations:

(a) hydrochloric acid with potassium hydroxide

(b) sulfuric acid, H_2SO_4, with sodium hydroxide

(c) nitric acid with calcium hydroxide, $Ca(OH)_2$

E. Challenge

8. Water acts as a base in reaction 3(a) but as an acid in $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$. What is the name for a substance that can do both, and how can water act both ways?

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